

On the Use of Electric Propulsion Systems for Future Space Missions
Compared to Chemical Propulsion Systems

Shing Hei Yeung

Abstract

Electric propulsion has been developing quickly in the recent decades. This paper aims to compare traditional chemical propulsion to electric propulsion and find out which system will be more prominent in future space missions. The history of space propulsion and exploration, and the thruster mechanisms of the two propulsion systems are discussed and compared to set a base for the comparison. The ability of the two propulsion systems in launching and travelling through space is first compared, including analysis of various maneuver techniques, the power supply penalty, and the durability of the propulsion systems. The efficiencies of the two propulsion systems are compared, along with the different propellants and their safety and environmental impacts. Quoting the father of modern rocketry Dr. Robert H. Goddard, "*It is difficult to say what is impossible, for the dream of yesterday is the hope of today and the reality of tomorrow.*", although chemical propulsion will still be significant in manned space missions and during launch, electric propulsion will slowly take over chemical propulsion in future unmanned space missions, gaining more significance in the field of spacecraft propulsion.

Glossary

EP	Electric Propulsion
I_{sp}	Specific Impulse
η	Efficiency
SEP	Solar Electric Propulsion
SERT	Space Electric Rocket Test
RP-1	Kerosene
LOX	Liquid oxygen
HET	Hall-effect thruster
NASA	National Aeronautics and Space Administration
KE	Kinetic energy
SOA	State-of-the-art
HiPEP	High Power Electric Propulsion
DS4G	Dual-Stage 4-Grid

Introduction

Chemical propulsion has long been a commonly utilised method of propulsion for different spacecrafts such as satellites, missiles, and rockets. Electric propulsion offers a very high efficiency whilst utilizing the propellant¹, meaning that it is most certainly a major competitor to chemical thrusters in deep space missions, as chemical thrusters have a relatively low efficiency in utilizing propellant compared to electric propulsion. Since the first satellite sputnik was launched into space, human space technological advance and exploration had been progressing nonstop, with electric propulsion being a more and more prominent in the past few decades, it is becoming an undeniably significant propulsion system that we must consider. This paper will be investigating and comparing the different aspects of electric and chemical propulsion.

¹ Jahn, Robert G. *Physics of Electric Propulsion*. Courier Corporation, 2012.

History

In the last few decades, both electric and chemical propulsion systems have been used fairly regularly, with electric propulsion especially taking off in the past decades, from SERT-1 in 1964 to Psyche in 2023².

Early in 1906, Goddard had speculated propulsion using electron acceleration with an electric grid. This idea was then further developed by Tsiolkovsky and Oberth, leading up to the SERT I and II tests in the 1960s to 70s.³ On the other hand, Chemical propulsion was an idea since the discovery of gunpowder. The first liquid propellant rocket was assembled and launched by Goddard in 1926,⁴ around 50 years before the first tests for electric propulsion. In 1969, NASA's Apollo 11 mission successfully landed mankind on the Moon.⁵ This depicts how much more mature chemical propulsion technology is compared to electric propulsion.

Since the first man-made spacecraft, Sputnik, was sent to space in 1957, human's exploration of space has not stopped. We have landed on the moon, launched thousands of spacecrafts into space, including satellites, rockets, and space shuttles. In recent decades, we have even started to venture further across the solar system and sent various rovers to Mars

² "Psyche Mission," blogs.nasa.gov, n.d., <https://blogs.nasa.gov/psyche/>.

³ Ronald Cybulski et al., "RESULTS from SERT I ION ROCKET FLIGHT TEST," accessed March 9, 2024, <https://ntrs.nasa.gov/api/citations/19650009681/downloads/19650009681.pdf>.

⁴ "Dr. Robert H. Goddard, American Rocketry Pioneer - NASA," NASA, accessed March 9, 2024, <https://www.nasa.gov/dr-robert-h-goddard-american-rocketry-pioneer/>.

⁵ NASA, "Apollo 11 - NASA," NASA, January 8, 2024, <https://www.nasa.gov/mission/apollo-11/>.

(Perseverance⁶ and Curiosity⁷), Mercury (Mariner 10⁸), and even probes to deep space (Voyager I⁹), away from the Sun.

Future Projections

Because our topic is comparing the usage of electric and chemical propulsion in future space missions, we must try to extrapolate what would be the main aim or goal for future space missions. From Nasa's documentation for budget in 2024, there is speculation for the next few years to be focusing on the Artemis Mission, which is aiming to send humans back on the Moon again since 1967, and there will be a Mars campaign which aims to send a manned crew to Mars.¹⁰ The Artemis project is rather short, with the trip to Moon only lasting a span of 10 days overall.¹¹ On the other hand, the Mars campaign will be a lot longer. Although there is not an official prediction for the length of the journey, we can use the travel time of the Perseverance rover as reference. The Martian rover took around 7 months to travel from Earth to Mars.¹²

⁶ NASA, "Mars 2020 Rover," Nasa.gov (Nasa, 2019), <https://mars.nasa.gov/mars2020>.

⁷ mars.nasa.gov, "Summary | Timeline – NASA's Mars Exploration Program," NASA's Mars Exploration Program, 2019, <https://mars.nasa.gov/msl/timeline/summary/>.

⁸ "Mariner 10 - NASA Science," science.nasa.gov, n.d., <https://science.nasa.gov/mission/mariner-10/>.

⁹ "Voyager - Mission Timeline," Nasa.gov, 2019, <https://voyager.jpl.nasa.gov/mission/timeline/#event-nasas-voyager-2-probe-enters-interstellar-space>.

¹⁰ NASA, "Budget Estimates," 2024, <https://www.nasa.gov/wp-content/uploads/2023/03/nasa-fy-2024-cj-v3.pdf>.

¹¹ Kathryn Hambleton, "NASA's First Flight with Crew Important Step on Long-Term Return to the Moon, Missions to Mars - NASA," NASA, August 27, 2018, <https://www.nasa.gov/missions/artemis/nasas-first-flight-with-crew-important-step-on-long-term-return-to-the-moon-missions-to-mars/>.

¹² mars.nasa.gov, "Trip to Mars," mars.nasa.gov, 2020, <https://mars.nasa.gov/mars2020/timeline/cruise/#:~:text=The%20cruise%20phase%20begins%20after>.

NASA is also rapidly developing an electric propulsion technology – Solar Electric Propulsion (SEP). The technology is due to finish developing in October 2028.¹³ The development of this technology shows that NASA believes that this technology will have potential in the future. According to an article published by NASA, SEP could be integrated with a nuclear propulsion system to speed up the journey to Mars.¹⁴ However, this would require the SEP technology to be fairly established and mature for it to work out. It might be too soon to conclude that SEP can be utilised in future journeys to Mars.

¹³ NASA, “Budget Estimates,” 2024, <https://www.nasa.gov/wp-content/uploads/2023/03/nasa-fy-2024-cj-v3.pdf>.

¹⁴ “Nuclear Propulsion Could Help Get Humans to Mars Faster - NASA,” February 12, 2021, <https://www.nasa.gov/solar-system/nuclear-propulsion-could-help-get-humans-to-mars-faster/#:~:text=Using%20low%20thrust%20efficiently%2C%20nuclear>.

Thruster mechanisms and models

As the title suggests, we will be comparing chemical and electric propulsion. Therefore, it is necessary for us to first define what is meant by each of these propulsion systems, the mechanisms of these systems and the types of thrusters that are used in these systems. This will be significant in our analysis and comparison later in the essay.

Chemical thrusters, or more commonly known as rocket engines, propel a spacecraft by releasing the stored chemical energy in the propellant. In a chemical propulsion spacecraft, the propellant undergoes combustion in a chamber. It reacts with the oxygen under high temperature to produce CO_2 and H_2O gases. These exhaust gases expand in the chamber and is focused into a nozzle and accelerated outwards. According to Newton's 3rd law, for every force, there is an equal force acting in the opposite direction. Hence, the exhaust gas being accelerated backwards will cause an equal force acting forwards on the spacecraft and propel the spacecraft.¹⁵

Electric thrusters, on the other hand, uses the electrical power to ionize the propellant. This is done by bombarding the propellant atoms with high-energy, high-speed electrons.¹⁶ These electrons can knock off 1 or more electrons from the atom to form positively charged ions. The charged ions are then accelerated by an electric field with the assistance of a magnetic field in the chamber. As the propellant is ejected from the thruster, it acts as a reaction mass in space, as the ions accelerate and exert a force backwards, an equal and opposite force is then exerted on the spacecraft, propelling it forwards. A neutralizer is located at the back of a

¹⁵ <https://www.grc.nasa.gov/www/k-12/airplane/rocket.html#:~:text=The%20combustion%20produces%20hot%20exhaust,or%20a%20propeller%20powered%20aircraft.>

¹⁶ "National Aeronautics and Space Administration NASA Facts Ion Propulsion," *NASA*, accessed March 3, 2024, https://www.nasa.gov/wp-content/uploads/2015/08/ionpropfact_sheet_ps-01628.pdf.

thruster to release electrons. These electrons can neutralise the ions expelled so the ions won't be attracted back to the negative charge within the spacecraft and offsetting the thrust.

Chemical propulsion systems have been long established and is a quite mature and developed field of technology. Over the years, a lot of different thruster designs have been tested out. However, most of them work on the same mechanism: combustion with oxygen supply, gas pushed out of nozzle, produces thrust. The main difference between these thrusters is mainly in terms of the different propellant combination and the different designs of the combustion chamber and nozzle. This is largely different to electric thrusters, where there are various different mechanisms to accelerate charged particles.

In EP systems, there are various types of thrusters. Major ones include gridded ion thrusters, Hall-effect thrusters (HET), resistojets and arcjets.¹⁷ All these thrusters rely on the same theory that an electric field can accelerate an ion to very high speeds, but they have different mechanisms. In this essay, I will be mainly focusing on gridded ion thrusters and HETs as they are the most commercially utilised ones, so they have the most data when used in a practical situation. Gridded ion thrusters directly shoot high speed electrons at propellant atoms whilst HETs use a magnetic field to limit electrons axial movement so released electrons from the cathode spin around the core at high speeds and collide onto released propellant atoms.¹⁸

¹⁷ K. Holste et al., "Ion Thrusters for Electric Propulsion: Scientific Issues Developing a Niche Technology into a Game Changer," *Review of Scientific Instruments* 91, no. 6 (June 1, 2020): 061101, <https://doi.org/10.1063/5.0010134>.

¹⁸ M. Martinez-Sanchez and J. E. Pollard, "Spacecraft Electric Propulsion-an Overview," *Journal of Propulsion and Power* 14, no. 5 (September 1998): 688–99, <https://doi.org/10.2514/2.5331>.

Overall, both chemical and electric propulsion are reaction propulsions, meaning they both rely on exerting some mass backwards to generate a reaction force forwards to propel the spacecraft. However, they use different types of propellant and different methods of accelerating the “mass” backwards. Electric propulsion can utilise electrostatic forces differently to accelerate masses backwards, meaning it could have different mechanisms suited for different uses. Conversely, chemical thrusters have much less different choices for thruster mechanism, but do still have quite a variety of thruster models to choose from.

Launch process:

In the following paragraphs, I will be comparing the ability for chemical and electric thrusters in various procedures which are common and essential in space travel. I will first be comparing their ability to accelerate to over the escape velocity at ground level. Then, their ability to travel and maneuver through space will be compared, which includes maneuver techniques, durability of spacecrafts and efficiencies.

Escaping Earth's gravitational field:

From Bonnerot's lecture notes on escape velocity¹⁹, Using the equation:

$$v_{esc} = \sqrt{\frac{2GM_p}{R_p}}$$

By substituting the suitable parameters of the Earth, we get that the escape velocity of Earth will be 11.2 km s^{-1} . Also, from the equation, we can see that the mass of the spacecraft has no effect on the escape velocity, meaning we do not have to take the masses of spacecrafts of different propulsion systems into consideration whilst comparing their suitability for future space missions.

To achieve a specific velocity from the surface of the Earth, the gravitational force of Earth and the air resistance of the air has to be overcome. This means that the net thrust of the

¹⁹ Bonnerot, Clément. "Escape Speed." Accessed November 13, 2023.
http://www.tapir.caltech.edu/~bonnerot/notes_lecture.pdf

thrusters of a spacecraft must at least overcome these two factors, neglecting factors like launch angle.

From data, the maximum thrust generated by an EP thruster is around 3N^{20} whilst the maximum thrust generated by a chemical thruster is around $6,700,000\text{N}^{21}$. By making a rough estimation, as the gravitational acceleration of Earth is 9.81ms^{-2} , for a 5000 kg EP spacecraft, the force of gravity will already be around $50,000\text{N}$, which is way over the maximum thrust that an EP thruster could possibly generate. Therefore, it is clearly impossible for an EP thruster to reach escape velocity with the current maturity of the technology. The sheer scale of difference (10^7) between the two generated thrusts means that if an EP spacecraft needs to generate a thrust closer to that of a chemical thruster, thousands of EP thrusters will be needed which is not practical at all.

Overall, this comparison shows that EP thrusters are not fit for launching a spacecraft and if an EP spacecraft needs to be launched into space, it must require the aid of a chemical spacecraft to send it into orbit before it can get on with its mission. This significantly affects the practicality of EP thrusters. Due to its reliant nature on chemical thrusters, an EP spacecraft must be accompanied by a chemical rocket in its launch, meaning 2 propulsion systems are utilised, heavily increase the cost of each launch of EP spacecrafts. However, considering the fuel-efficient nature of EP thrusters, EP spacecrafts are generally quite lightweight, so multiple EP spacecrafts could be launched in one chemical rocket launch, possibly offsetting the huge increase in cost.

²⁰ Guido Giammarinaro et al., “A Scaling Methodology for High-Power Magnetically Shielded Hall Thrusters,” *Journal of Electric Propulsion* 2, no. 1 (May 18, 2023), <https://doi.org/10.1007/s44205-023-00049-8>.

²¹ “F-1 Rocket Engine | National Air and Space Museum,” airandspace.si.edu, accessed March 3, 2024, https://airandspace.si.edu/collection-objects/rocket-engine-liquid-fuel-f-1/nasm_A19700271000.

Travelling through space:

In this part, I will mainly compare the ability for the thrusters to travel in short and long missions. First, I will define “short” missions as journeys where the spacecraft will not be transferred away from the orbit of the Earth. All other missions are then considered as long.

Short transits:

In short transits and journeys, as the spacecraft will not be leaving Earth’s orbit, it will likely be something like a satellite. We can compare the propulsion systems by 2 factors: the launch mass and orbital lifetime.

The initial launch mass of satellites using electric propulsion can be 45% lower due to its high I_{sp} . Therefore, with a lower launch mass, a less powerful launch vehicle can be used, hence further reducing the cost, and making electric propulsion a better choice. With a higher I_{sp} , electric thrusters can have a longer orbital lifetime than chemical thrusters with the same amount of propellant. A gridded ion thruster needs 1.4 kg of propellant to have an orbital lifetime of 19 years whilst a chemical thruster needs 2.3 kg of propellant to have an orbital lifetime of 19 years, needing around 64% more propellant.²² The longer the satellite orbital lifetime is, less satellites will be needed to be sent into space and less space rubbish will be produced.

Overall, from the two factors, EP satellites are definitely the better choice for short transits, with both a lower cost, a higher mass efficiency and a higher orbital lifetime.

²² Sara Miller et al., “Survey and Performance Evaluation of Small-Satellite Propulsion Technologies,” *Journal of Spacecraft and Rockets*, October 1, 2020, 1–10, <https://doi.org/10.2514/1.a34774>.

Long transits:

In long transits and journeys, the thruster ability of the propulsion systems can be determined by 3 major factors: orbit maneuver, durability, and its efficiency.

Maneuver techniques:

There are maneuvering techniques that can help alter a spacecraft's direction and velocity without using up as much propellant as propelling using only thrust from the thruster of a spacecraft. This chapter will be focusing on two techniques, gravity assist maneuver (slingshot) and the Oberth effect.

Gravity assist maneuver:

Also known as the slingshot²³ or swing-by technique, gravity assist maneuver is often used on spacecrafts which are on long missions. It alters the spacecraft's direction and velocity using the gravitational field of different astronomical objects. As a spacecraft approaches the periapsis of the orbit of a planet, it speeds up as it gets closer to the planet and experiences a larger gravitational force. As the spacecraft passes the periapsis, it is rotated by the orbit and then leaves the orbit of the planet. As it leaves, it is being pulled back by the planet's gravitational force and slows down to its original speed. However, as the planet is also orbiting the Sun, the planet itself also has a velocity. If when the spacecraft leaves the orbit of the planet, the velocity of the planet is parallel to the spacecraft's velocity. The spacecraft gains the velocity and gains speed from the planet.

²³ Bartlett, Albert A., and Charles W. Hord. "The Slingshot Effect: Explanation and Analogies." *The Physics Teacher* 23, no. 8 (November 1985): 466–73. <https://doi.org/10.1119/1.2341882>.

Generally, the velocity a spacecraft gains in a gravity assist slingshot can be calculated by the equation²⁴:

$$\Delta v = 2|v_{planet}| \cos \theta$$

The more acute the angle is, the greater the velocity gain will be. From the equation, we can see that thrust from the spacecraft does not really play a part in the velocity gain. Therefore, both chemical propulsion spacecrafts and EP spacecrafts can efficiently utilise this maneuver in long transits.

Oberth effect:

The Oberth effect is also known as the powered fly-by. It describes how a spacecraft can use the gravity well of a cosmological object to accelerate a spacecraft, as long as the rocket produce a thrust at the periapsis (the lowest and fastest point in its path)²⁵. Alternatively, retro-firing at this point can effectively slow down the spacecraft.²⁶ To understand which propulsion system can utilise this maneuver more effectively, we can use the kinetic energy equation:

$$KE = \frac{1}{2}mv^2 \quad (1)$$

Where m is the mass of the spacecraft and v is the initial velocity of the spacecraft.

²⁴ Adam Moran and John Mann, “Orbital Mechanics of Gravitational Slingshots Final Paper 15-424: Foundations of Cyber-Physical Systems,” 2016, <http://symbolaris.com/course/fcps16/projects/amoran.pdf>.

²⁵ Blanco, Philip Rodriguez, and Carl E. Mungan. “Rocket Propulsion, Classical Relativity, and the Oberth Effect.” *The Physics Teacher* 57, no. 7 (October 2019): 439–41. <https://doi.org/10.1119/1.5126818>.

²⁶ Blanco, Philip Rodriguez, and Carl E. Mungan. “Rocket Propulsion, Classical Relativity, and the Oberth Effect.” *The Physics Teacher* 57, no. 7 (October 2019): 439–41. <https://doi.org/10.1119/1.5126818>.

When the spacecraft exerts a thrust at the periapsis, it causes a change in velocity which we will call Δv . Then we can work out the change in kinetic energy by:

$$KE_f = \frac{1}{2} m(v + \Delta v)^2 = \frac{1}{2} m(v^2 + 2v \cdot \Delta v + \Delta v^2) = \frac{1}{2} mv^2 + mv \cdot \Delta v + \frac{1}{2} m(\Delta v)^2$$

$$\Delta KE = \frac{1}{2} mv^2 + mv \cdot \Delta v + \frac{1}{2} m(\Delta v)^2 - \frac{1}{2} mv^2 = mv \cdot \Delta v + \frac{1}{2} m(\Delta v)^2$$

$$\Delta KE = m(v \cdot \Delta v + \frac{1}{2} (\Delta v)^2)$$

As the mass of the spacecraft is the same for both terms, it is negligible in the comparison.

The two terms that determine the change in kinetic energy are $v \cdot \Delta v$ and $0.5(\Delta v)^2$. The second term $0.5(\Delta v)^2$ is the energy used for the maneuver, so the only “gain” in energy is $v \cdot \Delta v$. By comparing how electric and chemical thrusters are able to maximise this value, we can know which propulsion system is more optimized to use this maneuver. As there is no air resistance nor any form of resistive forces in space (except for gravitational fields), the spacecrafts can literally accelerate infinitely and there is no upper bound of velocity as long as thrust is applied, and the velocity is lower than the speed of light. The velocity purely depends on the length of travel. The further it has travelled, the more acceleration it will have done, and a higher velocity will be obtained.

In chemical propulsion systems, chemical thrusters can generate large amount of thrust in a short period of time, meaning a large acceleration and a large Δv . Therefore, the gain in KE $v \cdot \Delta v$ will be large. However, due to chemical thrusters’ low I_{sp} , they are seldom used for

deep space missions and are more commonly used for shorter trips, meaning they will not be able to reach higher velocities.

Conversely in EP systems, electric thrusters are incapable of producing a large thrust at all, with the maximum thrust being at around 3N. This means that there will only be a small acceleration and small Δv . Considering Voyager 1 and 2 are now travelling at over 15000ms^{-1} ²⁷, the ratio between the first term and the second term would be at 60000:1, meaning that as long as the spacecraft enters the maneuver at a high speed, a small thrust can allow the spacecraft to gain a lot of kinetic energy compared to the energy used for the maneuver.

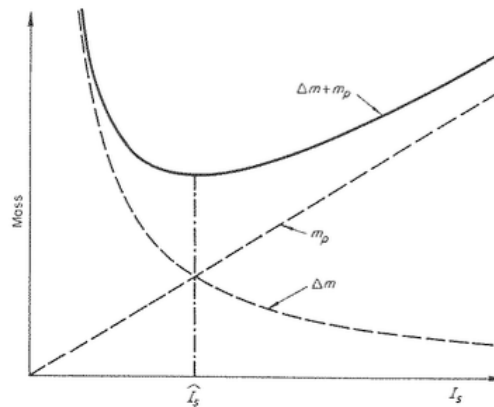
In conclusion, both electric and chemical propulsion systems are able to utilise the Oberth maneuver in their journey with high efficiency. However, due to the higher thrust of chemical propulsion systems, they will be able to gain more kinetic energy from this maneuver.

Power supply penalty:

The power supply penalty illustrates how the power supply of a thruster will alter its efficiency. In chemical thrusters, the propulsion only requires combustion of liquid oxygen and the propellant being sparked and brought to a high temperature to propel the spacecraft. Electric thrusters require electricity to be powered, meaning that in addition to the propellant, the battery which carry the electricity or even the solar arrays which are used to generate electricity in some spacecraft²⁸ will pose a significant penalty to the effectiveness of the thruster. Hence, the power supply penalty will be especially prominent in electric propulsion.

²⁷ NASA, "Voyager - Mission Status," Nasa.gov, 2012, <https://voyager.jpl.nasa.gov/mission/status/>.

²⁸ "Solar Electric Propulsion - NASA," NASA, accessed March 3, 2024, <https://www.nasa.gov/tdm/solar-electric-propulsion/>.



Figure(1) Graph of mass of spacecraft against I_{sp} ²⁹

There is a minimum point for the total mass of the spacecraft. This portrays how when the I_{sp} and the required power input increases, the mass of the power supply increases and causes the mass of the spacecraft to increase, hence with the same power and same thrust, less acceleration can be obtained. As more power supply is needed, the mass increases gradually and takes up more space. This significantly undermines the ability for EP thrusters to have higher thrusts as the required power input cannot be too high.

It could be argued that chemical thrusters also have most of their space and mass occupied by propellant and fuel³⁰. However, it is not specifically a power source so it should not be included in the comparison in this section.

As a result, chemical propulsion is a lot better than electric propulsion at minimising the power supply penalty, as the only power it requires is the power to operate the propellant flow system and overall system control.

²⁹ Robert G Jahn, *Physics of Electric Propulsion* (Courier Corporation, 2012).

³⁰ "Rocket Principles," web.mit.edu, accessed March 3, 2024, <https://web.mit.edu/16.00/www/aec/rocket.html#:~:text=For%20an%20typical%20rocket%2C%20the>.

Durability:

When discussing the ability of a propulsion system in long transit, the durability of the system is vital. Especially when a spacecraft is billions of kilometers from Earth, if the propulsion system breaks down, there is no easy way of maintaining and fixing it, basically meaning it is scrapped.

Chemical thrusters are simpler in structure. The fuel and oxidizer are transported to the thruster's combustion chamber to combust. The produced gas expands and is pushed out of the nozzle, hence propelling the spacecraft. However, in chemical propulsion systems, as combustion requires heat, the thermal energy inside the combustion chamber could cause damage to the different structures of the spacecraft, including the combustion chamber, system hardware and the high pressure could also lead to explosion.³¹

In contrast, EP thrusters are more complicated in its structure. There is a lot of electricity involved in the propulsion, including two grids producing an electric field to accelerate charged propellant ions and a tube at the back to release negatively charged ions to neutralize the propellant flow backwards.³² The increase in delicate mechanisms means a higher chance of getting damaged and lower durability. For example, if the neutralizer gets clogged and negatively charged particles are not able to be released³³, the exerted positive ions will be

³¹ D.F. Pennington, T. Man, and B. Persons, "Rocket Propulsion Hazard Summary: Safety Classification, Handling Experience and Application to Space Shuttle Payload," May 1977, <https://ntrs.nasa.gov/api/citations/19770018296/downloads/19770018296.pdf>.

³² Paul J. Wilbur, Vincent K. Rawlin, and J. R. Beattie, "Ion Thruster Development Trends and Status in the United States," *Journal of Propulsion and Power* 14, no. 5 (September 1998): 708–15, <https://doi.org/10.2514/2.5333>.

³³ "National Aeronautics and Space Administration NASA Facts Ion Propulsion," *NASA*, accessed March 3, 2024, https://www.nasa.gov/wp-content/uploads/2015/08/ionpropfact_sheet_ps-01628.pdf.

attracted to the negative grid of the thruster hence cancelling out the reaction force it exerted onto the spacecraft, resulting in a much lower efficiency of the thruster or even making the thruster completely useless. Moreover, the sputtering of all the accelerated particles will corrode the aerospace coating³⁴ of different components of the spacecraft, such as the solar arrays. This would significantly degrade the propulsion system and hence causing EP systems to be less durable.

Overall, both propulsion systems have flaws in their systems which leads to reduction in durability, but as long as the thruster is tested thoroughly and optimised before it is sent into space, it could minimise the chance of a defect in the propulsion system. Although EP thrusters have a more delicate and complex system and a less mature technology, chemical thrusters involve a lot of highly thermal reactions inside it which hugely increase the load and damage on the propulsion systems, and that is why traditional rocket boosters are all single-use and are usually just dropped into the atmosphere or into the ocean after usage.

³⁴ Shengfei Shang et al., “Sputtering Distribution of LIPS200 Ion Thruster Plume,” *Acta Astronautica* 160 (July 2019): 7–14, <https://doi.org/10.1016/j.actaastro.2019.04.008>.

Thruster efficiency:

Every type of thruster has different parameters to calculate efficiency. For example, for electric thrusters, there are mass utilization efficiency, electrical efficiency, and the discharge loss.³⁵ Chemical propulsion systems' efficiencies are determined by thermal efficiency and propulsive efficiency.

These efficiencies depend on a myriad of data and properties of the thrusters. Overall, we can express the thruster efficiency, in simple terms, as:

$$\eta = \frac{P_{jet}}{P_{electric/chemical}}$$

This is essentially a calculation on how well the thruster is able to convert the power that is put into the thruster into the actual power which is propelling the spacecraft. Considering the various parameters required to calculate the thruster efficiency, it is hard to determine if one thruster will have a higher or lower efficiency than the other. Therefore, there is not enough evidence to conclude which propulsion system has thrusters of better efficiencies.

³⁵ Dan M Goebel and Ira R Katz, *Fundamentals of Electric Propulsion*, 2008, <https://doi.org/10.1002/9780470436448>.

Specific Impulse

Specific impulse is defined as “the ratio of the thrust to the rate of propellant consumption”³⁶ and can be calculated using the equation:

$$I_{sp} = \frac{dp}{dm} = \frac{T}{\dot{m}} = v_{ex} \quad ^{37}$$

The specific impulse is the change in momentum per ejected mass, which is how efficiently a reaction engine generates thrust.³⁸ The consideration of specific impulse is particularly important whilst investigating a propulsion system’s ability in long, deep space missions. A high I_{sp} suggests a high mass efficiency, as the spacecraft can generate thrust for a way longer time with the same amount of propellant. This means that less propellant will be needed on the spacecraft, meaning either the total mass of a spacecraft can be reduced, or the payload of the spacecraft can be increased. With a higher I_{sp} , less propellant is needed, and more percentage of the spacecraft can now be allocated to the payload.

Chemical thrusters have a rather low I_{sp} (<400s). This means they do not use fuel as efficiently as EP thrusters. On average, most chemical propulsion spacecrafts have most of their mass and space being used to store the propellants for the thrusters. Over 90% of a

³⁶ Goebel, Dan M, and Ira R Katz. *Fundamentals of Electric Propulsion*, 2008 <https://doi.org/10.1002/9780470436448>.

³⁷ Holste, K., P. Dietz, S. Scharmann, K. Keil, T. Henning, D. Zschätzsch, M. Reitemeyer, et al. “Ion Thrusters for Electric Propulsion: Scientific Issues Developing a Niche Technology into a Game Changer.” *Review of Scientific Instruments* 91, no. 6 (June 1, 2020): 061101. <https://doi.org/10.1063/5.0010134>.

³⁸ Holste, K., P. Dietz, S. Scharmann, K. Keil, T. Henning, D. Zschätzsch, M. Reitemeyer, et al. “Ion Thrusters for Electric Propulsion: Scientific Issues Developing a Niche Technology into a Game Changer.” *Review of Scientific Instruments* 91, no. 6 (June 1, 2020): 061101. <https://doi.org/10.1063/5.0010134>.

chemical rocket is used for storing just fuel, 6% is used for structure and only 4% is used for payload.³⁹ The ratio of 1:24 between the payload and propellant is not really reasonable in a business point of view.

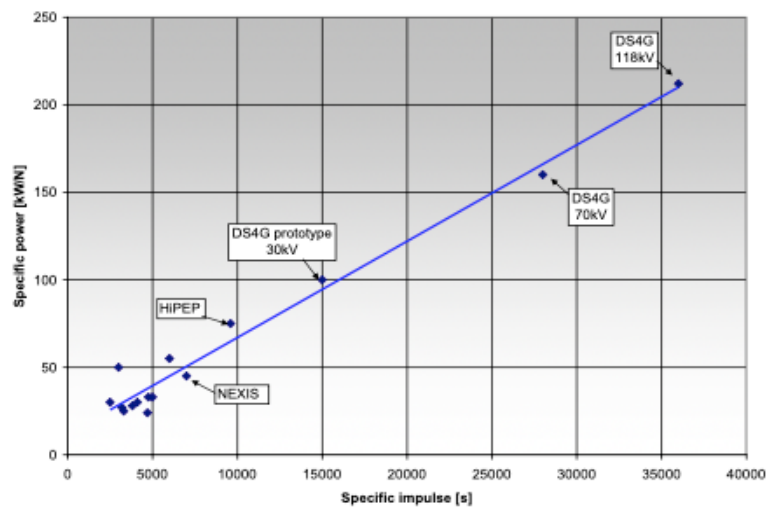
Electric propulsion systems specialise in enabling the EP thrusters to reach a very high I_{sp} ($I_{sp} > 2000s$) in order to help increase the thrust-to-weight ratio of thrusters.⁴⁰ This means that an electric thruster can generate the same amount of thrust of a chemical thruster but with less propellant. EP spacecrafts could also end up at a higher speed as less propellant is needed. According to equation (1), for any given amount of energy, the lower the mass, the higher the velocity.

Although electric propulsion systems do have a higher I_{sp} , as the I_{sp} further increase, the specific power of the spacecraft increases at a higher rate. The specific power (W/N) is the power required to generate a specific amount of thrust. From the following figure (2), for every 1000 second increase in the I_{sp} , the specific power increases by 5.5 kW/N. This might not be as significant in thrusters with a low thrust, because a low thrust will mean a lower increase in the required power input. However, for EP thrusters with a higher thrust at around 1N, it will definitely affect the performance of the thruster and affect the comparison. When the specific power increase, the overall required power will also increase, hence meaning more electrical power will need to be stored on the spacecraft. A larger storage of power

³⁹ “Rocket Principles,” web.mit.edu, accessed March 3, 2024, <https://web.mit.edu/16.00/www/aec/rocket.html#:~:text=For%20an%20typical%20rocket%2C%20the>.

⁴⁰ Angelo Genovese and Nadim Maraqtan, “Advanced Electric Propulsion Concepts for Fast Missions to the Outer Solar System and Beyond,” *Journal of the British Interplanetary Society* 76, no. 4 (June 14, 2023): 114–21, <https://doi.org/10.59332/jbis-076-04-0114>.

means that the mass of the spacecraft will increase, undermining the advantages of reduced propellant mass from the increase in I_{sp} .

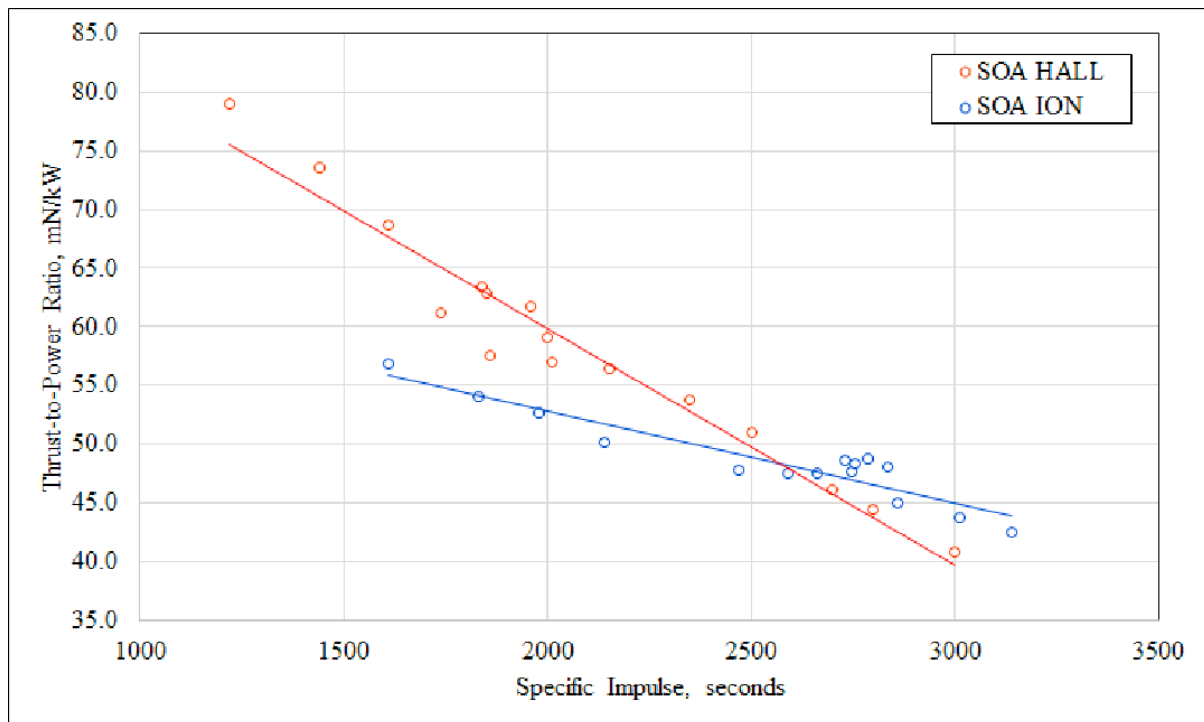


Figure(2), Graph of Specific Power (kW/N) against specific impulse ⁴¹

In order to minimize the effect of this phenomenon, either the I_{sp} and specific power needs to be optimized, or a power supply with a lower specific mass needs to be used.

⁴¹ Angelo Genovese and Nadim Marqten, "Advanced Electric Propulsion Concepts for Fast Missions to the Outer Solar System and Beyond," *Journal of the British Interplanetary Society* 76, no. 4 (June 14, 2023): 114–21, <https://doi.org/10.59332/jbis-076-04-0114>.

Relationship with efficiency:



Figure(3) Graph of Thrust-to-Power against Specific Impulse ⁴²

In EP systems, the I_{sp} of the spacecraft is closely related to its thrust-to-power ratio. Figure (3) is a graph of the thrust-to-power ratio against the specific impulse of gridded ion thrusters and hall effect thrusters respectively. From the graph, 2 conclusions could be made:

- i) When the I_{sp} is lower at around the range of 1250-2500s, the Hall-effect annular thruster can generate a higher thrust with the same amount of power provided compared to a gridded ion thruster, meaning Hall-effect thrusters are optimized to maximise the thrust-to-power of the spacecraft.
- ii) When the I_{sp} is higher at over 2700s or above, the Hall-effect thruster's graph went under the graph of the gridded ion thruster, suggesting the gridded ion thruster now has a higher thrust-to-power ratio. From the graph, the gridded ion

⁴² Michael J. Patterson et al., "High Thrust-To-Power Annular Engine Technology," 51st AIAA/SAE/ASEE Joint Propulsion Conference, July 23, 2015, <https://doi.org/10.2514/6.2015-3719>.

thruster also has a much shallower gradient, so as the I_{sp} increase, the thrust-to-power will drop slowly, suggesting they are more optimized towards maximizing the I_{sp} , which is the propellant efficiency of the thruster.

Overall, EP systems can cater for both high efficiency (high I_{sp}) and high thrust-to-power depending on the situation and usage, whilst chemical systems are more orientated to focus on high thrust and has rather low I_{sp} . Therefore, EP system is definitely a better choice for any deep space missions where a short duration of travel is not necessary as its high I_{sp} can allow it to produce thrust for a lot longer than chemical propulsion with the same amount of propellant.

Propellant, safety, and their environmental impacts

Every propulsion system requires a propellant. Chemical propulsion uses the exhaust gases from combustion to produce a thrust and utilise the reaction force to propel the spacecraft. Similarly, electric thrusters also create thrust using the reaction force. However, the exhaust is produced by accelerating charged propellant ions backwards.

The technology of chemical propulsion is well-established and there are hundreds of different propellant combinations to use in chemical thrusters. This section will be focusing on bipropellant thrusters. The propellant is stored in either liquid or solid, and there is always a need for an oxidiser to provide oxygen for combustion in space. A few commonly used propellants include RP-1(kerosene), LH2(liquid hydrogen) and UDMH(hydrazine).⁴³

RP-1 is a hydrocarbon that is used to fuel the internal combustion taking place in the spacecraft. A major problem for combustion is that it is an exothermic reaction and so it would lead to a high heat. A high heat combined with the fact that RP-1 is a highly flammable chemical means that any mistakes in the structure or design flaw in the spacecraft will lead to a possible explosion.

Chemical thrusters use combustion, therefore a lot of different pollutants will be released into the Earth's atmosphere during launch. According to research conducted by Dallas et al., the major emission pollutants from LOX/RP-1 propellant are carbon oxides, nitrogen oxides and

⁴³ J.A. Dallas et al., "The Environmental Impact of Emissions from Space Launches: A Comprehensive Review," *Journal of Cleaner Production* 255, no. 1 (May 2020): 120209, <https://doi.org/10.1016/j.jclepro.2020.120209>.

soot. These pollutants could deplete the ozone layer, cause acid rains and smog, and would also enhance the greenhouse effect and speed up global warming.⁴⁴

Xenon gas is used as the propellant for most EP systems nowadays. Xenon gas has a mass number of 131.4 and is a group 8 element. This indicates that it is a noble gas, which has the property of not chemically reactive as it is stable as a monatomic molecule. One advantage of Xenon gas as a propellant over RP-1 in chemical thrusters is that it is not combustible⁴⁵, and it is not flammable like RP-1. This significantly reduces the safety hazard during a rocket assembly and rocket launch.

Mercury and Cesium has been used as propellants at the beginning of the development of EP thrusters⁴⁶. However, mercury is toxic to humans and cesium's high surface tension coats the insulators, causing technical difficulty in maintain the system, so they are no longer used.

Fullerene (C₆₀) has been suggested to be used as a propellant⁴⁷ for EP thrusters due to its high atomic mass of 720 amu. Compared to Xenon's 131.4, it has nearly 6 times as much mass, meaning each fullerene will generate 6 times more thrust than each Xenon atom. This would be especially useful for higher thrust EP spacecrafts. However, a major disadvantage of

⁴⁴ J.A. Dallas et al., "The Environmental Impact of Emissions from Space Launches: A Comprehensive Review," *Journal of Cleaner Production* 255, no. 1 (May 2020): 120209, <https://doi.org/10.1016/j.jclepro.2020.120209>.

⁴⁵ PubChem (n.d.). *Xenon*. [online] pubchem.ncbi.nlm.nih.gov. Available at: <https://pubchem.ncbi.nlm.nih.gov/compound/xenon>.

⁴⁶ "SUMMARY on the REPORT DEVELOPMENT of the SERT I SPACECRAFT," February 1, 1966, <https://ntrs.nasa.gov/api/citations/19660008296/downloads/19660008296.pdf>.

⁴⁷ Paul J. Wilbur, Vincent K. Rawlin, and J. R. Beattie, "Ion Thruster Development Trends and Status in the United States," *Journal of Propulsion and Power* 14, no. 5 (September 1998): 708–15, <https://doi.org/10.2514/2.5333>.

fullerene is that it could be easily fragmented when bombarded with electrons which lowers the performance of the discharge chamber.

Aromatic hydrocarbons are another type of propellant which has been suggested from Borrfors' research.⁴⁸ They have concluded that due to their similar mass, lower cost, and possibly higher ionization efficiency, they are potential propellants in the future.

Overall, as shown from above, a lot of propellants are not particularly suitable for EP propulsion, resulting in the selection of propellant for EP spacecrafts being quite narrow, without many choices except for possibly aromatic hydrocarbons, which are still in a theoretical stage, and the inert gases, especially Xenon. This might cause shortages in these gases and could also lead to an increase in value for these propellants and an increase in the cost of using an EP spacecraft.

⁴⁸ André Nyberg Borrfors et al., "Aromatic Hydrocarbons as Molecular Propellants for Electric Propulsion Thrusters," *Journal of Electric Propulsion* 2, no. 1 (November 16, 2023), <https://doi.org/10.1007/s44205-023-00059-6>.

Solution – for now:

Both electric and chemical propulsion have their own advantages and disadvantages in space travel, neither can be defined as a complete “winner” over the other. Instead of trying to perfect one of these propulsion systems, multimode spacecraft propulsion is emerging as a propulsion system with great potential, with the ability to take the capabilities of both systems into one. Multimode propulsion is essentially multiple propulsion systems merged into one single propulsion system.⁴⁹ With the use of a multimode, the best of both chemical and electric propulsion can be merged into one propulsion system, allowing for high propellant efficiency when travelling through space and high thrust during launch. However, the complexity of such a system and current technology immaturity means that this idea will probably require a few more decades before it might be more widely used in future space missions.

⁴⁹ Joshua L. Rovey et al., “Review of Multimode Space Propulsion,” *Progress in Aerospace Sciences* 118 (October 1, 2020), <https://doi.org/10.1016/j.paerosci.2020.100627>.

Conclusion:

After comparisons between chemical and electric propulsion in a myriad of different abilities and properties, I could conclude that electric thrusters will possibly be the main thruster in use for future unmanned space missions and spacecrafts as the technology continues to advance. They have high I_{sp} values and have low masses and reduced amount of propellant, allowing them to have high orbital lifetimes as satellites and maintain constant thrust over a very long period of time in deep space missions, potentially accelerating spacecrafts to speeds above 15000ms^{-1} . It can utilise different gravitational maneuver techniques efficiently, meaning it can make use of other planets on its deep space journey.

However, due to its low thrust nature, unless there are some major engineering breakthroughs which will allow electric thrusters to generate high thrust, chemical thrusters will not be completely replaced by electric thrusters due to its excellent ability to generate a very large thrust. This is portrayed especially clearly in the launch stage of the spacecraft. Electric thrusters currently are completely reliant on chemical thrusters in escaping the Earth's gravity. In addition, manned missions will also still rely on chemical propulsion as its high thrust can ensure a short transit time, meaning minimal daily necessities will be needed on the spacecraft, and reducing the risk of any catastrophes.

Electric propulsion is going to be prominent in future unmanned exploration missions, but chemical propulsion will still be used in manned space missions and during the launch of a spacecraft.

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